

EKEDGE, Turkmenistan

WMO: 383880

Lat: **41.033N**Lon: **57.767E**Elev: **62**StdP: **100.58**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-13.6	-21.1	0.6	-16.0	-17.8	0.8	-10.9	8.2	1.7	7.7	2.1	1.2	50	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	42.1	21.9	40.6	21.2	39.1	20.6	24.7	37.7	23.3	36.9	22.1	35.9	3.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	15.0	33.8	18.5	13.4	31.3	16.8	12.1	30.3	75.1	37.8	69.5	37.0	64.8	36.0	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.6	6.3	-20.8	44.4	6.0	1.3	-25.2	45.3	-28.7	46.1	-32.1	46.8	-36.4	47.7		
(5)			-21.2	25.2	5.8	2.4	-25.3	26.9	-28.8	28.3	-32.0	29.6	-36.3	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	-1.5	1.3	8.8	16.3	23.4	28.5	30.9	28.9	21.9	13.8	5.8	0.4
	DBStd	12.31	6.06	6.01	5.28	5.15	4.06	3.33	3.20	3.61	4.18	5.15	5.30	4.88
	HDD10.0	1159	356	247	86	8	0	0	0	0	0	22	143	297
	HDD18.3	2596	613	478	299	97	8	0	0	0	14	156	376	554
	CDD10.0	2970	0	3	48	198	415	556	647	587	358	140	17	1
	CDD18.3	1367	0	0	3	37	165	307	389	328	122	16	1	0
	CDH23.3	20603	0	0	38	459	2143	4732	6346	5029	1624	224	7	0
	CDH26.7	12568	0	0	9	179	1086	2981	4234	3229	784	65	1	0
Wind	WSAvg	2.4	2.1	2.5	2.8	2.8	2.7	2.7	2.5	2.4	2.3	2.2	2.1	2.2
Precipitation	PrecAvg	112	10	13	17	19	14	6	2	1	3	7	10	8
	PrecMax	207	29	28	54	65	69	31	8	19	26	29	46	36
	PrecMin	34	0	1	1	0	1	0	0	0	0	0	0	0
	PrecStd	46	7	7	12	15	16	9	3	4	6	8	13	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	19.3	27.8	35.0	38.6	42.4	45.0	43.7	38.5	31.6	23.9	15.4
		MCWB	8.7	9.7	15.0	17.7	20.5	22.2	23.2	21.6	19.9	17.5	14.5	8.9
	2%	DB	9.9	14.8	23.4	31.3	36.2	40.6	42.7	41.6	35.8	28.5	19.2	11.8
		MCWB	6.0	8.2	12.9	16.1	19.3	21.3	22.1	21.5	18.9	15.8	12.1	6.7
	5%	DB	7.7	11.8	20.2	28.2	34.3	39.2	41.0	39.8	33.5	25.8	16.6	9.2
		MCWB	4.5	6.6	11.4	15.3	18.3	20.6	21.3	20.9	17.7	14.1	10.3	5.8
	10%	DB	5.6	9.0	17.2	25.5	32.4	37.5	39.4	38.1	31.4	23.3	14.1	7.1
		MCWB	3.3	5.1	9.9	14.2	17.8	19.8	20.9	20.2	17.1	13.2	8.9	4.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	10.5	15.6	18.7	23.7	27.1	26.9	25.1	22.7	19.0	15.5	9.3
		MCDB	12.9	17.8	26.1	30.8	35.1	39.6	40.6	36.9	35.9	29.1	22.5	13.9
	2%	WB	6.4	8.4	13.5	17.1	21.0	24.1	25.1	23.2	20.4	16.5	12.8	7.6
		MCDB	9.4	14.0	22.0	28.7	32.7	36.3	38.8	37.8	31.3	26.3	18.4	11.1
	5%	WB	4.8	6.9	11.9	16.0	19.6	22.6	23.6	22.0	18.8	15.0	11.0	6.0
		MCDB	7.1	11.3	19.3	26.5	31.0	35.6	37.6	37.1	31.6	24.0	15.4	8.8
	10%	WB	3.4	5.4	10.4	14.7	18.5	21.3	22.2	20.9	17.6	13.7	9.3	4.6
		MCDB	5.4	8.6	16.7	24.3	29.8	34.5	36.3	36.3	30.2	22.0	13.5	7.1
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.9	11.8	12.6	13.9	14.7	14.4	15.5	15.6	14.5	11.5	8.7
		MCDBR	11.2	14.2	15.5	16.5	16.4	17.2	16.1	17.5	17.8	17.5	15.1	12.0
		MCWBR	7.5	8.7	7.9	7.0	7.0	6.7	6.4	7.1	7.9	9.0	9.3	8.3
	5% WB	MCDBR	10.3	12.9	14.4	15.1	15.0	15.0	15.0	15.9	16.5	15.5	13.9	11.0
MCWBR		7.3	8.3	7.7	7.2	7.7	8.1	7.8	7.6	8.6	8.7	8.9	8.0	
Clear-Sky Solar Irradiance	taub	0.402	0.438	0.495	0.553	0.550	0.552	0.548	0.511	0.479	0.471	0.426	0.398	
	taud	2.001	1.912	1.818	1.728	1.772	1.776	1.800	1.864	1.904	1.896	1.983	2.013	
	Ebn at Noon	697	736	741	731	747	746	744	757	748	689	662	661	
	Edh at Noon	122	155	191	224	219	218	211	193	173	155	123	112	
All-Sky Solar Radiation	RadAvg	1.79	2.69	4.19	5.47	6.74	7.66	7.30	6.76	5.47	3.74	2.34	1.61	
	RadStd	0.22	0.36	0.29	0.41	0.33	0.29	0.27	0.25	0.21	0.23	0.21	0.21	